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Long-Term Survivorship After Chimeric Antigen Receptor T-Cell Therapy for Hematologic Malignancies

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ABSTRACT

Clinical trials and real-world evidence have established the benefit of chimeric antigen receptor (CAR) T-cell therapy to reduce progression of hematologic malignancies and, in some settings, increase overall survival. However, recent studies have highlighted that patients in long-term remission remain at risk of morbidity and nonrelapse mortality, primarily because of infections and subsequent myeloid malignancies. In addition, these patients may also be survivors of chemotherapy, radiation, and/or stem-cell transplant, with broad survivorship needs. Here, we provide a roadmap for the care of patients in long-term remission after CAR-T-cell therapies and discuss avenues for future research.

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A ROADMAP TO SURVIVORSHIP CARE

Chimeric antigen receptor (CAR) T-cell therapy has transformed the treatment landscape for patients with relapsed and/or refractory hematologic malignancies.[1-5] Much attention has focused on early adverse effects of CAR-T-cell therapy, such as cytokine release syndrome (CRS) and neurotoxicity, which is warranted as they affect morbidity, mortality, and health care utilization.[6,7] However, it is important to note that many issues with CAR-T-cell therapy occur well beyond this initial acute period. First, the most common cause of death after CAR-T-cell therapy is relapse of the original malignancy, and therefore, improving the long-term antitumor efficacy of CAR-T cells is critically needed. Second, long-term survivors in remission remain at ongoing risk of nonrelapse mortality (NRM). A recent report from the US Lymphoma Consortium described a 5-year NRM rate of 16%, with over half of the NRM events occurring beyond 2 years.[1] Conversely, late relapses were rare. Meta-analyses, retrospective studies, and large registries are remarkably concordant: the majority of NRM among CAR-T-cell survivors is due to infection, with substantial late NRM (5%-10% of patients) because of secondary malignancies, including both myeloid malignancies and solid tumors.[1,8,9]

In addition to mortality events, long-term morbidity and patient-reported adverse outcomes are common.[10-13] Overall, there is paucity of evidence-based guidelines for screening, prevention, and management among CAR-T-cell recipients. While survivorship considerations after CAR T-cell therapy share certain themes with those observed after hematopoietic stem-cell (HSC) transplantation (HCT)—such as immune dysfunction, cytopenias, and infection risk—CAR T-cell therapy is biologically and clinically distinct. Given these differences and acknowledging current gaps in CAR-specific data, we outline a pragmatic roadmap to survivorship care (Fig 1; Table 1).

INFECTION

Infections are a major complication of CAR-T-cell therapy, occurring in 20%-60% of CAR-T recipients.[24] While most patients recover, infections are the most important cause of NRM, representing half of all events.[25] We found that 17% of patients with lymphoma required hospitalization or intravenous antibiotics between 6 and 12 months after CAR-T-cell therapy, and 10% between 1 and 2 years after infusion.[1] Patients treated with CAR-T-cell therapy have underlying immune dysregulation because of inflammation and immunosuppression associated with their active cancer and the effects of previous treatments. Immunity is further disrupted by lymphodepleting chemotherapy, the use of treatments to suppress cytokine inflammation, and the on-target effects of CAR-T cells, leading to B-cell aplasia.[26] Over the long term, many patients subsequently experience prolonged cytopenias and poor immune reconstitution of B- and T-lymphocytes.

Bacterial Infections

Bacterial infections account for the majority of early infections post–CAR-T; however, data on late infection are limited.[27] Careful attention should be paid, in particular, to sinus and pulmonary infections, with a low threshold for antibiotic coverage at symptom onset or in patients with nonresolving syndromes. For patients experiencing recurrent or nonresolving infections, immunoglobulin replacement is recommended.[28] For patients receiving plasma cell–directed therapies (ie, B-cell maturation antigen [BCMA]-targeted CAR T cells for myeloma), prophylactic immunoglobulin replacement is recommended. Overall, a high index of clinical

	Intervention	Year 1	Year 2	>3 years
Hematology	Disease-specific staging	Follow disease-specific restaging assessments based on risks of early and late relapse		
	CBC	CBC at routine visits to assess for cytopenia		
	NGS with or without BMBX	If ongoing, new, or worsening cytopenia, assess for CHIP and/or secondary myeloid neoplasms (Fig 2)		
Oncology	Secondary malignancy screening	Age-appropriate cancer screenings. For recipients of high-dose chest radiation, recommend screening mammogram at age 30 years or >10 years post-radiation, whichever is latest		
Immunology and infections	Infections	Maintain increased vigilance for infections with a low threshold to initiate treatment		
	Vaccines	Emphasize respiratory pathogen and shingles vaccines starting 3-6 months post–CAR T; defer live vaccines until ≥1 year post–CAR T. Vaccines more effective once B- and T-cell recovery occurs.		
	Antimicrobial prophylaxis	Antibacterial: Fluoroquinolone for prolonged neutropenia Antiviral: Acyclovir/valacyclovir ≥ 6 months. Anti–HBc-positive patients should receive hepatitis prophylaxis for least 12 months Pneumocystis: TMP-SMX (or equivalent) ≥ 6 months or until CD4 > 200/μL. Fungal: Consider mold prophylaxis for prolonged neutropenia.		
	Immunoglobulin replacement	Monitor IgG levels; target IgG ≥400 mg/dL in those with recurrent sinopulmonary infections Some patients (ie, recipients of BCMA CAR T) may benefit from prophylaxis		
Neurology	Neurology examination	Monitor BCMA CAR T recipients for movement disorders and cranial nerve palsies.		
Cardiology	History	Assess cardiovascular risk (family history, lifestyle, comorbidities, hx. anthracycline, chest radiation). Refer patients with reduced LVEF or high cardiac risk post–CAR T to cardiology for ongoing monitoring		
Sexual health and fertility	Sexual function	Discuss sexual function with patients at 3 months post–CAR T and annually thereafter		
	Family planning	Discuss safe sex and contraception, especially with AYA patients; refer those desiring pregnancy to fertility specialists.		
Gastrointestinal	Gastrointestinal toxicity	Evaluate diarrhea for Immune-effector cell–associated enterocolitis (BCMA CAR T recipients) and infection		
Special populations	Geriatric assessments	Consider geriatric assessments prior to CAR T in older adults to evaluate physical and functional status, cognition, polypharmacy, nutrition, and social support		
	Pediatric survivorship	Age-specific issues such as endocrine/growth, fertility, psychosocial, and family survivorship		

FIG 1. Roadmap to survivorship care for CAR-T-cell recipients. This figure outlines a multidisciplinary approach to survivorship and late-effect monitoring after CD19- or BCMA-directed CAR T-cell therapy. Interventions are organized by system domain and year of follow-up, reflecting distinct temporal patterns of risk. Depending on the model of care and patient preferences, some of these activities may be completed at the CAR-T treatment center and some at the referring community practice, whereas others may involve primary care providers. BCMA, B-cell maturation antigen; BMBX, bone marrow biopsy; CAR-T, chimeric antigen receptor T-cell therapy; CHIP, clonal hematopoiesis of indeterminate potential; LVEF, left ventricular ejection fraction; NGS, next generation sequencing.

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TABLE 1. Selected Late Toxicities of CD19- and BCMA-Directed CAR T Cells

Domain	CD19-Targeted CAR T	BCMA-Targeted CAR T	Recommendations
Late infections	Mixed pattern with a shift toward respiratory viral infections[14]	Respiratory viral infections account for a majority of late infections[15]	Low threshold for viral PCR testing; IVIG may reduce infection risk. Consider bacterial superinfections in severe or nonresolving viral cases
Hypogammaglobulinemia and immune reconstitution	B-cell aplasia may persist beyond the first year, most recover by 3 years[16]	B-cell recovery median 70 days; plasma cell aplasia approximately 7 months[17]	IVIG for recurrent sinopulmonary or severe infections for goal IgG level >400 mg/dL. Prophylaxis warranted in the first year after BCMA-directed CAR T-cell therapy
Prolonged cytopenias	Prolonged cytopenias common.[18,19]	Prolonged cytopenias common; thrombocytopenia more common than after CD19[20]	Often ICAHT; workup and management outlined in Figure 2
Secondary malignancies	Incidence of therapy-related myeloid neoplasms 3%-5% within 1-2 years; T-cell lymphoma rare[21]	Therapy-related myeloid neoplasms observed within 1-2 years; T-cell lymphoma rare[22]	Maintain a low threshold for bone marrow biopsy with cytopenias or baseline high-risk CHIP clones (eg, *TP53* mutations). Continue lifelong surveillance for secondary malignancies with age-appropriate cancer screening and monitor for new or worsening cytopenias
Neurologic late events	Delayed ICANS that onsets beyond 2 weeks is rare	Non-ICANS neurologic complications (Parkinsonism, cranial nerve palsies) median onset at 3-4 weeks among cilta-cel recipients[23]	Close monitoring of neurologic symptoms with early referral to neurology warranted for BCMA CAR T recipients
Diarrhea	No reported cases of IEC-EC, recommend workup of secondary causes	IEC-enterocolitis may occur up to several months after CAR T-cell therapy	Upper and lower endoscopy with biopsy for cases of nonresolving or otherwise unexplained diarrhea. Imaging may reveal colitis. Apart from IEC-EC, common infectious and viral reactivation causes may occur

Abbreviations: BCMA, B-cell maturation antigen; CAR-T, chimeric antigen receptor T-cell therapy; CHIP, clonal hematopoiesis of indeterminate potential; ICAHT, Immune Effector Cell–Associated Hematotoxicity; ICANS, immune effector cell–associated neurotoxicity syndrome; IEC-EC, immune effector cell–associated enterocolitis; IVIG, intravenous immune globulin; PCR, polymerase chain reaction.

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suspicion should be maintained, with prompt diagnostic evaluation and early initiation of empiric antimicrobial therapy.

Viral Infections

Respiratory viral infections are the leading cause of late infections in part due to delayed B and T-cell immune reconstitution.[29] In one study of BCMA CAR-T recipients, the 1-year cumulative incidence of respiratory infections was 22%, with a majority of cases described as mild/moderate.[15] While most resolve, a subset of patients can experience serious complications including respiratory failure, secondary bacterial and/or fungal infections, and even death.

SARS-CoV-19 Infection

During the COVID-19 pandemic, CAR-T recipients had disproportionally high rates of severe SARS-CoV-2[30] infection because of profound B-cell aplasia and impaired humoral immunity.[31,32] In a multicenter prospective observational study of the safety and immunogenicity of SARS-CoV-2 vaccination after cellular therapy, vaccine-related adverse events were uncommon.[33] However, efficacy was muted by the fact that in CAR-T patients with ongoing B-cell aplasia, binding and neutralizing antibodies did not increase after vaccination. Meaningfully, vaccine-specific T-cell responses were observed after the second vaccine.[33] Thus, the Infectious Diseases Society of America (IDSA) guidelines recommend initiating COVID-19 vaccines beginning at ≥3 months after CAR-T.[34]

Although the severity of SARS-CoV-2 infection has lessened over time, cellular therapy recipients are still at risk for poor outcomes. In CAR-T-cell survivors who develop symptomatic SARS-CoV-2 infection, we have a low threshold for antiviral treatment and immunoglobulin replacement.

Nonrespiratory Viral Reactivations

Viral reactivations are rarely reported in recipients of CD19- and BCMA-directed CAR-T-cell therapy.[24,35,36] Extrapolating from the HCT literature, the American Society of Transplantation and Cellular Therapy (ASTCT) guidelines

recommend measurement of infectious disease markers including HIV and hepatitis B and C as part of the initial CAR-T workup.[28] Cases of fatal reactivation have been described in chronic hepatitis B carriers, and these patients require prophylaxis for at least 1 year after CAR-T infusion and close monitoring upon cessation of prophylaxis (assuming B-cell recovery).[28] Patients who have had hepatitis C previously eradicated do not require specific monitoring. HIV reactivation is rare, and caution is needed to interpret the results of viral testing after treatment with products that include HIV sequences within the CAR transgene.[37]

Cytomegalovirus (CMV) reactivation is observed in 20%-60% of CMV seropositive patients, with most infections occurring within 2-6 weeks of CAR-T.[38-41] Risk factors include BCMA-targeted therapy and receipt of ≥3 days of corticosteroids and/or anakinra.[38,39] The clinical significance of CMV viremia in this setting is not well-elucidated with only rare reports of CMV end organ disease.[41] ASTCT guidelines recommend weekly CMV surveillance in the first 2-6 weeks for high-risk patients including CMV seropositive patients treated with BCMA-directed therapies and those requiring ≥3 days of corticosteroids.[28] Treatment is decided on a case-by-case basis. There are no data supporting CMV prophylaxis (ie, letermovir), but this warrants further evaluation.

Antimicrobial Prophylaxis

Recommendations for antimicrobial prophylaxis mainly rely on retrospective experiences and extrapolations from the transplant literature.[28,36] Assessment of pre–CAR-T risk factors, such as previous allogeneic HCT and prolonged cytopenias, should be considered so that monitoring and intervention can be personalized. We recommend antibacterial prophylaxis with fluoroquinolones in patients with prolonged severe neutropenia. For herpes simplex virus and varicella zoster virus (VZV) prevention, we use acyclovir or valacyclovir for at least 6 months and often longer in patients with prolonged B-cell aplasia who do not generate humoral responses to the zoster vaccine. To prevent *Pneumocystis jirovecii*, we recommend trimethoprim/sulfamethoxazole (TMP/SMX) or atovaquone until at least 6 months and the CD4 count is >200 cells/mm^3.[28] While invasive fungal infections are rare, they can be life-threatening, and antimold azoles are recommended for those with risk factors such as prolonged cytopenias, prolonged steroid use, or history of invasive fungal infections.

B-Cell Aplasia, Hypogammaglobulinemia, and Vaccination

Hypogammaglobulinemia and immune responses to vaccination may depend on the CAR-T-cell target, CAR-T-cell persistence, and B- and T-cell immune reconstitution.[14] For example, CD19 is expressed on normal B cells but typically not on long-lived plasma cells, which express BCMA. Thus, patients who receive CD19 CAR-T-cell therapy do not develop new humoral responses to vaccination while B-cell aplasia is present, but do maintain some pre-existing antibody-mediated immunity in their long-lived plasma cells.[42,43] Among patients with lymphoma, recovery of normal B cells typically occurs in 50%-70% of patients by 1 year after therapy, with a greater proportion recovering after 3 years.[1,16,35,44] By contrast, BCMA-targeted CAR-T depletes plasma cells which are critical for maintaining existing humoral immunity. This may account for increased late infections in this patient population; however, responses to new vaccination can still occur.[17,45] Higher rates of seroconversion to influenza and SARS-CoV-2 vaccine have been observed among BCMA CAR-T recipients as compared with CD19 CAR-T recipients.[45,46]

Experience to date indicates that vaccination is safe in CAR-T-cell survivors; data are less clear about their efficacy to prevent infection, and thus, they remain premature to enforce a universal vaccination schedule post–CAR-T. With this in mind, revaccination with nonlive vaccines is reasonable to consider beginning 3-6 months postinfusion and is more likely to be effective in patients demonstrating B- and T-cell reconstitution.[34] Because the majority of clinically significant infections after CAR-T-cell therapy involve respiratory pathogens, priority for initial vaccination efforts includes influenza, SARS-CoV-2 infection, and pneumococcus, as well as VZV because of the risk of shingles while immunocompromised. RSV vaccines are recently approved and are of interest as a respiratory pathogen, but data are currently lacking in the CAR-T population. Other vaccines that are typically undertaken in the post-HCT population (eg, tetanus or hepatitis B) are recommended in guidelines but are of less clear benefit, and some centers pursue individualized strategies such as titer-guided revaccination or other risk stratifications. While harms have not been described in CAR-T patients specifically, live vaccines are typically avoided until at least 1 year after cell infusion in those with demonstrated immune reconstitution and no intravenous immune globulin (IVIG) use in the past 8 months.[28,47]

A key risk factor for respiratory infections is hypogammaglobulinemia. For plasma cell–directed CAR-T cells (ie, BCMA), expert consensus is to provide universal prophylaxis in the first year until IgG levels recover above 400 mg/dL.[48] For CD19 CAR-T-cell therapy survivors, replacement for IgG levels ≤400 mg/dL is recommended for those with recurrent infections.[28,49] In general, available data do not inform which patients should receive immunoglobulin prophylaxis in the absence of infections and which patients only require immunoglobulin replacement in the setting of recurrent or nonresolving infections. In addition, practice varies as to the duration of IVIG therapy after initiation since some patients do not recover B-cell production even many years after CAR-T-cell therapy, and indefinite monthly IVIG may be impractical or costly. In long-term survivors with ongoing hypogammaglobulinemia, one may consider providing IVIG for 3 months and then return to observation, continuing replacement longer if symptoms have not resolved or if

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infections were severe or recurrent. Prospective studies are ongoing to identify patients who would most benefit from IVIG after CAR-T-cell therapy (ClinicalTrials.gov identifier: NCT05952804).

CYTOPENIAS

Cytopenias are common after CAR-T-cell therapy, and European collaborative groups have developed the Immune Effector Cell–Associated Hematotoxicity (ICAHT) grading system, including specific criteria for neutropenia (N-ICAHT) and thrombocytopenia (T-ICAHT).[50,51] This system incorporates both depth and duration of cytopenias and distinguish between early (0-30 days) and late (>30 days) hematotoxicity. Severe N-ICAHT is associated with prolonged neutropenia, multilineage cytopenias, transfusion dependence, higher infection rates, increased NRM, and worse overall survival (OS), and T-ICAHT has similarly been linked to transfusion burden, bleeding risk, and inferior survival outcomes.[50,51]

Early Cytopenias (Days 1-30)

Most patients have already received multiple lines of chemotherapy and often present with baseline cytopenias. Subsequently, lymphodepleting regimens—typically fludarabine and cyclophosphamide—are highly myelosuppressive, inducing grade 4 cytopenia in more than half of patients.

Late Cytopenias (after Day 30)

Prolonged and recurrent cytopenias are increasingly recognized as significant complications of CAR-T-cell therapy. Severe cytopenia affects 50%-80% of patients, with approximately 5%-10% of patients continuing to have significant cytopenias up to two years after therapy.[1,35,44,52] Persistent severe neutropenia, particularly an aplastic phenotype, is associated with risk of severe or fatal infection.[51,53-55] Emerging data suggest that CAR-T cells are causative of the cytopenias, with mechanisms including cytokine-induced injury to myeloid progenitors and early apoptosis of mature neutrophils.[20,56,57]

Late cytopenias may reflect a multifactorial pathophysiology. Bone marrow disease burden is a key predictor, with patients with ALL experiencing more severe and prolonged cytopenias than those with lymphoma because of extensive marrow involvement.[58] In addition, post–CAR-T-specific mechanisms—such as cytokine-mediated marrow injury, emergence of suppressive clonal T-cell populations (eg, large granular lymphocytes [LGLs]), and expansion of mutated myeloid clones—further impair hematopoietic recovery.[51,53,54,59,60]

Among these mechanisms, clonal hematopoiesis of indeterminate potential (CHIP) is an emerging contributor. CHIP refers to myeloid gene mutations with a variant allele frequency of ≥2% in individuals without overt myeloid malignancies. CHIP is found in 30%-50% of recipients before lymphodepletion and up to 80% post–CAR-T infusion.[61-63] Certain mutations—particularly in *DNMT3A*, *TET2*, and *ASXL1*—have been associated with increased rates of early immune toxicities, such as increased rates of immune effector cell–associated neurotoxicity syndrome (ICANS).[64,65] In some settings, particularly when combined with clinical variables such as inflammation and preexisting cytopenias, CHIP has been associated with hematologic toxicity, whereas in other studies, the association has been more limited.[21,56,61,63,64,66] Refined risk stratification models—incorporating CHIP burden, mutation type, and inflammatory markers—may better identify patients at risk of hematologic vulnerability after CAR-T-cell therapy. As will be discussed in later sections, CHIP may also place patients at risk of secondary myeloid malignancies.

Predictive Models for Hematotoxicity

The CAR-HT score stratifies risk of post–CAR-T hematologic toxicity using baseline laboratory tests (platelets, hemoglobin, ANC) and inflammatory markers (CRP, ferritin). High scores predict prolonged cytopenias, infections, NRM, and inferior progression-free survival/OS. Its utility has been validated in lymphoma and myeloma cohorts, with an online calculator available for clinical use.[67] In B-acute lymphoblastic leukemia (B-ALL) , where marrow infiltration is nearly universal, CAR-HT showed limited predictive value. The ALL-HT score, which substitutes ferritin with marrow disease burden, better discriminates risk.[58]

Workup for Post–CAR-T Cytopenia

Identifying the cause of cytopenia after CAR-T-cell therapy is often complex.[18] Initial evaluation should review myelosuppressive drugs, nutritional status, infection risk, and disease. For instance, patients on TMP/SMX may benefit from alternative prophylactic agents. A rapid ferritin rise may suggest immune effector cell–associated hemophagocytic syndrome (IEC-HS). The concomitant presence of circulating LGL may warrant a trial of corticosteroids. Viral testing—targeted based on clinical context—may include CMV, Epstein-Barr virus, parvovirus B19, and adenovirus. Further workup may include bone marrow evaluation and staging studies to determine cancer remission status. Most prolonged cytopenias are attributable to ICAHT, typically presenting as marrow hypocellularity without another identifiable cause, and resolve with time.

Management of Cytopenias

Management centers on close monitoring and supportive care, including transfusions, antimicrobials, and growth factors (Fig 2). Despite these approaches, most available treatments remain empiric and often ineffective.

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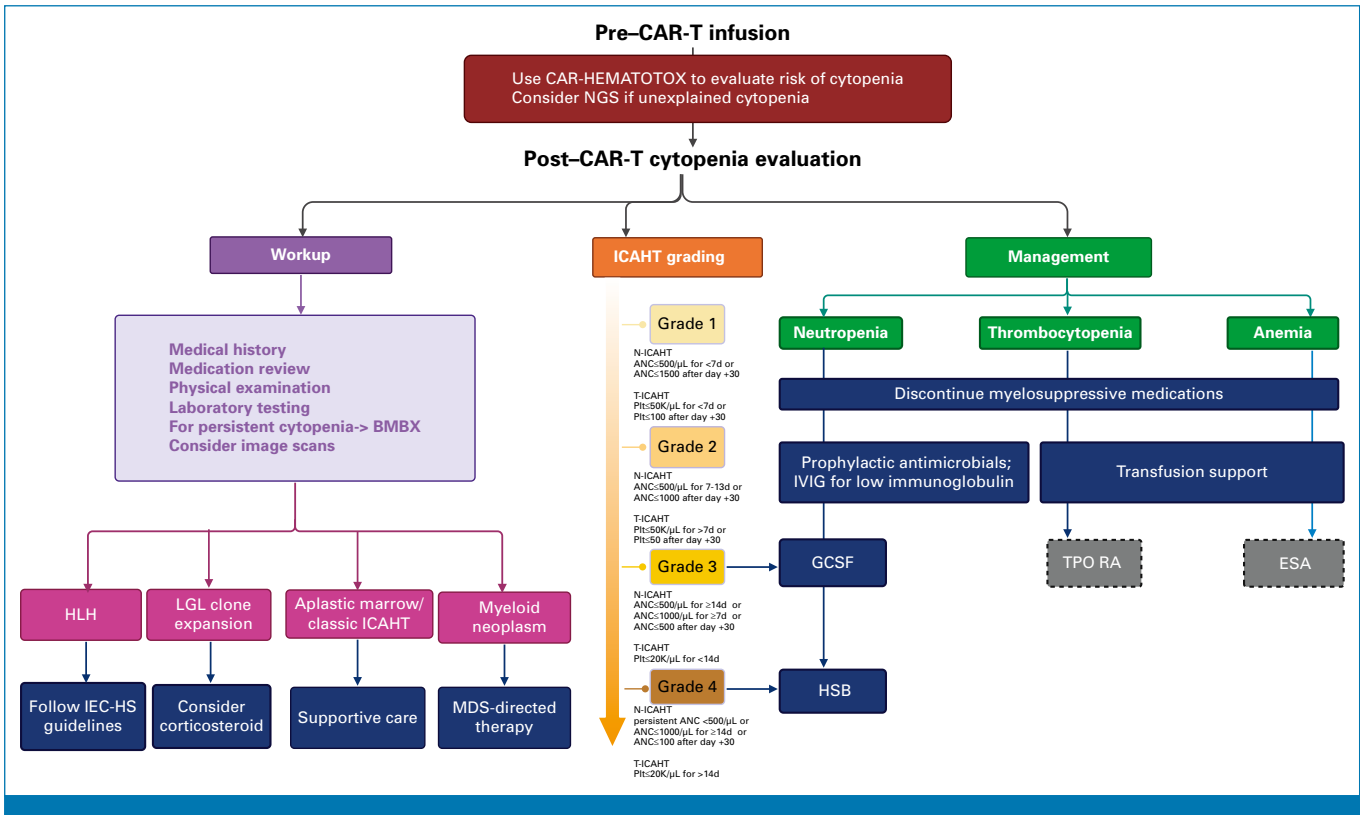

FIG 2. Recommended workup and management of post–CAR-T cytopenia (the dashed line indicates limited efficacy). BMBX, bone marrow biopsy; CAR-T, chimeric antigen receptor T-cell therapy; ESA, erythroid-stimulating agent; GCSF, granulocyte colony-stimulating factor; HLH, hemophagocytic lymphohistiocytosis; HSB, hematopoietic stem cell boost; ICAHT, Immune Effector Cell–Associated Hematotoxicity; IEC-HS, immune effector cell–associated hemophagocytic syndrome; IVIG, intravenous immune globulin; LGL, large granular lymphocyte; MDS, myelodysplastic syndrome; TPO RA, thrombopoietin receptor agonist.

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Granulocyte Colony-Stimulating Factor Use

Granulocyte colony-stimulating factor (G-CSF) is a mainstay of supportive care for late neutropenia. However, many severe cases of ICAHT are refractory to G-CSF, limiting benefit. Granulocyte-macrophage colony stimulating factor is avoided because of its potential to worsen inflammatory complications.[68,69]

Immunosuppressive Therapy

For severe cases of cytopenia in which flow cytometry reveals the expansion of oligoclonal T cells or LGLs, a trial of low-dose steroids may be considered to suppress possible cytokine-mediated cytopenia.[59,70]

HSC Boost

Autologous CD34$^+$ HSC boosts appear to be one of the few effective interventions for restoring hematopoiesis in patients with prolonged cytopenia after CAR-T-cell therapy.[71,72] However, availability of stored stem cells is often limited, especially as CAR-T moves earlier in treatment algorithms. The European Hematology Association /European Society for Blood and Marrow Transplant panel recommends HSC infusion without conditioning for patients with severe ICAHT (grade ≥3) beyond day 14, if a product is available and the patient is G-CSF–refractory. There are reports that stem-cell boosts may exacerbate inflammation, and caution is warranted with low initial stem-cell doses recommended if boost is absolutely necessary in patients with previous IEC-HS.[73]

Thrombopoietin Receptor Agonists and Erythroid-Stimulating Agents

Supportive care remains the mainstay for post–CAR-T thrombocytopenia and anemia. Data on thrombopoietin receptor agonists (eg, eltrombopag, romiplostim) are limited, and their optimal dosing and overall utility remain uncertain.[74,75]

Allogeneic HCT

The rationale for use in this setting is extrapolated from aplastic anemia, where transplant can be curative. Importantly, most cases of ICAHT demonstrate spontaneous recovery over time, and the substantial risk of transplant-related morbidity and mortality generally outweighs the uncertain benefit.

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SECONDARY MALIGNANCIES

Long-term data indicate a 4%-16% incidence of secondary malignancies after CAR-T-cell therapy.[52,76,77] While these rates are comparable with those seen with conventional treatments for hematologic cancers, the underlying mechanisms remain incompletely understood. Contributing factors may include previous genotoxic therapies, CAR-T–induced inflammation, prolonged B-cell aplasia, immunosuppression, and pre-existing CHIP.

Secondary Solid Tumors

A large postmarketing study observed a 15.2% cumulative incidence over 5 years, with a median time to onset of 26.4 months.[78,79] These rates appear to be consistent with expectations based on age and previous therapies, and it remains unclear whether CAR-T-cell therapy independently increases solid tumor risk.

Therapy-Related Myeloid Neoplasms

Among secondary malignancies, therapy-related myeloid neoplasm (t-MN)—including myelodysplastic syndrome (MDS) and AML—is of particular concern. One study reported a 9% incidence of t-MN, with 5% progressing to high-grade MDS or AML requiring treatment.[21] Another study observed a lower overall incidence of MDS/AML (1.9%) post–BCMA CAR-T.[65] CHIP has also been linked to elevated t-MN risk: one study reported a 24-month cumulative incidence of 19% in CHIP-positive patients versus 4.2% in CHIP-negative, with shared mutations between baseline CH and subsequent t-MNs in some cases.[64] However, other studies found t-MN to be rare, highlighting heterogeneity across disease settings and cohorts.[19,63] Critically, *TP53*-mutated CHIP clones have demonstrated strong leukemogenic potential, often showing marked post–CAR-T expansion and portending poor outcomes (median OS 7 months).[21,22,60] By contrast, other CHIP mutations did not display similar behavior. These findings highlight the high leukemic potential of *TP53*-mutated CHIP, suggesting that CAR-T-cell therapy may create selective pressure favoring the expansion of high-risk clones. One challenge is the lack of effective therapies to prevent CHIP progression into tMNs, and the CAR-T patient population that benefits from screening, monitoring, or other interventions remains to be defined.

T-Cell Lymphoma

Rare cases of secondary T-cell lymphoma (TCL) have been reported after CAR-T-cell therapy, with 22 cases identified among approximately 8,000 patients in the US Food and Drug Administration Adverse Event Reporting System.[80] Proposed mechanisms include pre-existing lymphoid mutations (lymphoid CHIP) and, in rare instances, vector integration from CAR constructs.[79,81,82] However, not all post–CAR-T TCLs harbor the CAR transgene, and vector integration remains a research tool with limited clinical applicability. Currently, these cases are managed using standard protocols for de novo TCL although detailed molecular characterization studies may support targeted treatment approaches for individual patients.[83]

Survivorship Considerations for Secondary Malignancy

The emerging data on secondary malignancies—both solid tumors and hematologic neoplasms—highlight the need for proactive survivorship strategies. Adherence to age-appropriate cancer screening, including dermatologic screening, per national guidelines should be a core component of long-term care. For individuals with prolonged cytopenias, poor immune reconstitution, or known CHIP, more frequent and individualized monitoring may be warranted. As we gain deeper insights into the interplay between CAR-T-cell therapy, CHIP, and long-term immune dysfunction, personalized surveillance frameworks will be essential for mitigating late toxicities and optimizing outcomes for CAR-T-cell recipients.

ORGAN DYSFUNCTION

While multiorgan dysfunction in the setting of CRS is well-described after CAR-T, delayed organ dysfunction is uncommon and typically occurs in the setting of infections and/or as a sequela of previous therapy. Here, we describe more commonly observed organ toxicities after CAR-T and propose guidelines for monitoring.

Cardiac Toxicity

Retrospective studies report that approximately 10%-20% of CAR-T recipients develop early cardiopulmonary adverse events during CRS, mainly driven by troponin elevations and arrhythmias.[84-89] Little is known about long-term cardiac outcomes. Extrapolating from the HCT literature, we recommend assessing for and managing cardiac risks annually starting at 6 months after CAR-T.[90] Furthermore, CHIP is a known predictor of cardiovascular disease in autologous HCT recipients, where it is associated with higher rates of heart failure, coronary artery disease, stroke, and NRM, and further study is needed in long-term survivors of CAR-T-cell therapy because of the high baseline CHIP burden and post–CAR-T clonal expansions.[91,92]

Neurologic Toxicity

ICANS is common after CAR-T-cell therapy; however, it almost always starts within the first 2 weeks after infusion.[7,93] Most cases of myelopathy are also reported within the first 2 weeks after CAR-T but could occur later, manifesting with subacute lower extremity weakness, gait disturbance, and urinary dysfunction, and often associated with spinal cord MRI abnormalities and CSF pleocytosis.[94,95] Unique to BCMA-directed CAR-T-cell therapy are non-ICANS neurotoxicities (NINTs), presenting as Parkinsonism or cranial nerve palsies.[23] Hypokinetic movement disorders occur in 1%-5% of patients treated with ciltacabtagene autoleucel (cilta-cel) with the onset ranging from 11 to 108 days (median 26.5 days).[23] Cranial nerve palsies occur more frequently, with a

similar time to onset.[96] Optimal management is unknown, but corticosteroids, intravenous immunoglobulins, intrathecal chemotherapy, and/or cyclophosphamide are all used.[96,97] The likelihood of late onset is poorly described, but cases have been reported as far as 2.5 years post–CAR-T, underscoring the importance of long-term follow-up.[98] We recommend comprehensive neurologic examinations during routine follow-up visits in addition to patient counseling.[99]

GI Toxicity

Immune effector cell–associated enterocolitis (IEC-EC), characterized by copious diarrhea beginning a median of 3 months after CAR-T therapy (range, 1-7 months), is a key emerging toxicity of BCMA-directed CAR-T.[100] This entity is characterized by infiltration of CAR-T cells into the gastrointestinal tract and has been associated with gut perforation and sepsis. Optimal management is unclear, with some cases resolved by corticosteroids or anti-TNF or JAK inhibition, whereas others have required cyclophosphamide or other attempts to ablate the CAR. New diarrhea occurring within months of BCMA CAR-T therapy should be investigated thoroughly with consideration given to the diagnosis of IEC-EC.

PSYCHOSOCIAL IMPACT AND PATIENT-REPORTED OUTCOMES

Patient-reported outcomes (PROs) are key to understanding the lived experiences of patients.[10,11] Among 120 patients with

TABLE 2. Key Questions and Proposed Approaches for Survivorship Post–CAR-T

Domain	Key Questions	Proposed Approaches
Survivorship risks	Which risks are universal to CAR T-cell therapy and which risks are variable based on the number and type of previous treatments, CAR T-cell products and targets, and type of disease treated?	Collect data relevant to NRM and survivorship consistently across clinical trials, registries, and retrospective studies to allow comparisons
Care delivery models	What is the optimal care model for long-term survivorship, especially for patients treated at referral centers? How can we improve coordination between academic centers and community oncologists? Can digital tools assist with long-term survivorship?	Develop hybrid survivorship models that combine academic and community care Implement standardized handoffs and survivorship care plans after CAR T-cell therapy Study remote monitoring using wearable devices and ePRO platforms Integrate real-time alerts into survivorship pathways to enable early intervention
Infection prevention	Can closer monitoring of infections, immunoglobulin supplementation, or revaccination reduce NRM?	Establish protocols for long-term antimicrobial prophylaxis, immunoglobulin supplementation, and revaccination Study late NRM as an end point for prospective interventional trials
Risk stratification	Can we develop a precision survivorship model that integrates genomic profiles, inflammatory markers, clinical variables, and patient-reported outcomes to guide individualized long-term care?	Develop dedicated survivorship programs for CAR-T recipients Validate tools in multicenter cohorts to guide surveillance intensity
Cardiometabolic health	What are the long-term risks of cardiovascular disease and metabolic syndrome after CAR T-cell therapy?	Evaluate the incidence of heart failure, coronary disease, and metabolic syndrome post–CAR T Evaluate the long-term benefit of cardiac risk reduction strategies (ie, statins)
Organ dysfunction	What are the prevalence and trajectory of long-term pulmonary, hepatic, and renal toxicities after CAR T-cell therapy? Can early detection strategies reduce chronic organ dysfunction?	Conduct longitudinal studies tracking pulmonary, hepatic, and renal function beyond 1 year Develop early detection and intervention protocols to mitigate chronic dysfunction
Secondary malignancies	What are the incidence and mechanisms of t-MN, T-cell lymphomas, and solid tumors? Can early screening interventions improve outcomes of secondary malignancies?	Screen for CHIP pretreatment and monitor high-risk mutations (eg, *TP53*) over time Implement age-appropriate cancer screening and educate patients on second malignancy symptoms Design prospective studies to define causality and optimal surveillance intervals
Patient-reported outcomes	What is the long-term impact of CAR T-cell therapy on cognition, fatigue, fertility, sexual health, and financial toxicity? How do PROs evolve beyond 1 year, and what is their association with clinical outcomes?	Expand collection of longitudinal PROs beyond 1 year using validated tools Assess impacts on cognition, fatigue, sexual health, fertility, and financial burden.—Link PROs to clinical outcomes to support value-based care
Special populations	Which aspects of CAR-T survivorship are universal, and which are unique to elderly, AYA, and pediatric populations?	Acknowledge differences and identify unique survivorship needs based on age and frailty

Abbreviations: AYA, adolescents and young adult; CAR-T, chimeric antigen receptor T-cell therapy; CHIP, clonal hematopoiesis of indeterminate potential; NRM, nonrelapse mortality; PRO, patient-reported outcome; t-MN, therapy-related myeloid neoplasm.

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DLBCL treated with CAR-T at our institution, quality-of-life scores improved or remained similar through the first year of treatment.[101] Similar findings have been reported from randomized trials.[102] In a single-institution experience, preexisting mental health disorders were associated with increased risk of toxicity after CAR-T but did not affect efficacy.[103] Several knowledge gaps remain, including the impact on fertility, ability to return to work post-CAR-T and its financial implications, and trajectory of PROs beyond the first few months of treatment. We endorse the TCT position to incorporate the Patient-Reported Outcomes Measurement Information System instrument in clinical practice and clinical trials.

CONSIDERATIONS FOR SPECIAL POPULATIONS

Older Adults

Lymphoma and multiple myeloma predominantly affect older adults. Real-world data have demonstrated that older patients who receive CAR-T have similar response rates as younger patients.[1,2,5] However, among recipients with lymphoma of CAR-T-cell therapy, patients older than 60 years had similar OS but higher rates of NRM, underscoring the particular importance of survivorship care for older adults.[1] A growing body of evidence supports that frailty is a better determinant of outcomes than age.[104,105] Further study is needed for optimal baseline risk stratification. In addition, serial comprehensive geriatric assessments at baseline, 6 months, and 1 year may help to optimize outcomes among older adults.[90,106]

Adolescents and Young Adults

Survivorship is an important issue for adolescents and young adults (AYAs) age 15-39 years with hematologic malignancies. Among this unique patient population, declines in sexual health and fertility are key concerns.[107] In one longitudinal study, over half of AYA cancer survivors reported problematic sexual functioning at every assessment time point, with sexual dysfunction worsening over time.[108] While little is known about fertility after CAR-T, live births of healthy infants are case-reported.[109] To address the current knowledge gaps, an ongoing study (ClinicalTrials.gov identifier: NCT07173322) is enrolling survivors to collect detailed patient-reported reproductive histories, including fertility preservation decisions, pregnancy outcomes, and psychosocial impacts. We recommend early referral to reproductive endocrinology in patients interested in fertility preservation at time of initial consult and establishment of a programmatic AYA strategy.

Pediatrics

Pediatric survivors have unique needs, and many survivors have a history of both CAR-T-cell therapy and allo-HCT.[110] Programmatic long-term monitoring for CAR-T-cell–associated mutagenesis, secondary malignancies, relapse, fertility, growth, and development is particularly critical for patients who received these therapies at a young age and may become true long-term survivors.[111]

RESEARCH AND FUTURE DIRECTIONS

Structured survivorship programs after HCT improve long-term outcomes, but similar data are lacking for CAR-T survivors.[90] To outline the challenges ahead, Table 2 highlights several key unanswered questions spanning care delivery models, strategies to reduce NRM, cardiometabolic risk, and the long-term burden of treatment-related toxicities. Notably, standardized and transparent reporting of NRM and late adverse events in prospective CAR-T-cell trials remains inconsistent, limiting our ability to accurately quantify and mitigate survivorship risks. To accelerate progress, coordinated multi-institutional efforts, national registries, and prospective survivorship trials are needed. Work is needed toward a precision survivorship framework that integrates clinical characteristics, genomic profiles, inflammatory markers, and PROs to support individualized care and improve survivorship outcomes.

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Long-Term Survivorship After Chimeric Antigen Receptor T-Cell Therapy for Hematologic Malignancies

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